

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **n-Octane**  
 Product Description: **n-Octane**  
 Cat No. : C12937  
 CAS No 111-65-9  
 Molecular Formula C<sub>8</sub> H<sub>18</sub>

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
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 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Flammable liquids                                  | Category 2 (H225) |
| Aspiration Toxicity                                | Category 1 (H304) |
| Skin Corrosion/Irritation                          | Category 2 (H315) |
| Specific target organ toxicity - (single exposure) | Category 3 (H336) |
| Acute aquatic toxicity                             | Category 1 (H400) |
| Chronic aquatic toxicity                           | Category 1 (H410) |

**Label Elements**


# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

## Signal Word

## Danger

### Hazard Statements

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H336 - May cause drowsiness or dizziness  
H410 - Very toxic to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No   | Weight % |
|-----------|----------|----------|
| Octane    | 111-65-9 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

**Inhalation** Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. Risk of serious damage to the lungs (by aspiration).

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Special hazards arising from the substance or mixture**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Evacuate personnel to safe areas. Remove all sources of ignition. Use personal protective equipment as required. Ensure adequate ventilation. Take precautionary measures against static discharges.

**Environmental precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods and Material for Containment and Cleaning Up**

Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

### Conditions for Safe Storage, Including any Incompatibilities

Flammables area. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV    | OSHA PEL                                                                                                                                                                            |
|-----------|----------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Octane    |          | TWA: 300 ppm | (Vacated) TWA: 300 ppm<br>(Vacated) TWA: 1450 mg/m <sup>3</sup><br>(Vacated) STEL: 375 ppm<br>(Vacated) STEL: 1800 mg/m <sup>3</sup><br>TWA: 500 ppm<br>TWA: 2350 mg/m <sup>3</sup> |

| Component | European Union | The United Kingdom | Germany                                                                                                                                                                                                                                                                                                                            |
|-----------|----------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Octane    |                |                    | TWA: 500 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 2400 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 500 ppm (8 Stunden). MAK except Trimethylpentane isomers<br>TWA: 2400 mg/m <sup>3</sup> (8 Stunden). MAK except Trimethylpentane isomers<br>Höhepunkt: 1000 ppm<br>Höhepunkt: 4800 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                       |
|----------------|-----------------------|
| Appearance     | Colorless             |
| Physical State | Liquid                |
| Odor           | Petroleum distillates |
| Odor Threshold | No data available     |
| pH             | Not applicable        |

|                     |                               |                                   |
|---------------------|-------------------------------|-----------------------------------|
| Melting Point/Range | -57 °C / -70.6 °F             |                                   |
| Softening Point     | No data available             |                                   |
| Boiling Point/Range | 125 - 127 °C / 257 - 260.6 °F | @ 760 mmHg                        |
| Flash Point         | 13 °C / 55.4 °F               | Method - No information available |

|                          |                                  |        |
|--------------------------|----------------------------------|--------|
| Evaporation Rate         | 0.6 (Butyl Acetate = 1.0)        |        |
| Flammability (solid,gas) | Not applicable                   | Liquid |
| Explosion Limits         | Lower 0.8 Vol%<br>Upper 6.5 Vol% |        |

|                              |                          |                       |
|------------------------------|--------------------------|-----------------------|
| Vapor Pressure               | 14 mbar @ 20 °C          |                       |
| Vapor Density                | 3.9                      | (Air = 1.0)           |
| Specific Gravity / Density   | 0.708                    |                       |
| Bulk Density                 | Not applicable           | Liquid                |
| Water Solubility             | Insoluble                | practically insoluble |
| Solubility in other solvents | No information available |                       |

### Partition Coefficient (n-octanol/water)

|           |         |
|-----------|---------|
| Component | log Pow |
| Octane    | 5.18    |

|                           |                          |                                             |
|---------------------------|--------------------------|---------------------------------------------|
| Autoignition Temperature  | 220 °C / 428 °F          |                                             |
| Decomposition Temperature | No data available        |                                             |
| Viscosity                 | 0.55 mPa.s at 20 °C      |                                             |
| Explosive Properties      |                          | Vapors may form explosive mixtures with air |
| Oxidizing Properties      | No information available |                                             |

# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

Molecular Formula C<sub>8</sub> H<sub>18</sub>  
Molecular Weight 114.23

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** Hazardous polymerization does not occur.  
**Hazardous Reactions** None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Temperatures above 200°C.

### Incompatible Materials

Strong oxidizing agents.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

(a) acute toxicity;  
Oral No data available  
Dermal No data available  
Inhalation No data available

| Component | LD50 Oral     | LD50 Dermal      | LC50 Inhalation               |
|-----------|---------------|------------------|-------------------------------|
| Octane    | >5 g/kg (Rat) | >2 g/kg (Rabbit) | LC50 > 24.88 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available

# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

|                                                   |                                                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin</b>                                       | No data available                                                                                                                   |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                   |
| <b>(f) carcinogenicity;</b>                       | No data available<br>There are no known carcinogenic chemicals in this product                                                      |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                   |
| <b>(h) STOT-single exposure;</b>                  | Category 3                                                                                                                          |
| <b>Results / Target organs</b>                    | Central nervous system (CNS).                                                                                                       |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                   |
| <b>Target Organs</b>                              | No information available.                                                                                                           |
| <b>(j) aspiration hazard;</b>                     | Category 1                                                                                                                          |
| <b>Symptoms / effects, both acute and delayed</b> | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.                |
| <b>Endocrine Disrupting Properties</b>            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish | Water Flea                          | Freshwater Algae | Microtox               |
|-----------|-----------------|-------------------------------------|------------------|------------------------|
| Octane    |                 | EC50: = 0.38 mg/L, 48h (water flea) |                  | EC50 = 890 mg/L 30 min |

### **Persistence and degradability**

**Persistence** May persist, based on information available.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** May have some potential to bioaccumulate

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Octane    | 5.18    | No data available             |

### **Mobility in soil**

The product is insoluble and floats on water. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Spillage unlikely to penetrate soil. Will likely be mobile in the environment due to its volatility. Is not likely mobile in the environment due its low water solubility. Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not let this chemical enter the environment Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1262  
Hazard Class 3  
Packing Group II  
Proper Shipping Name OCTANES

### Road and Rail Transport

UN-No UN1262  
Hazard Class 3  
Packing Group II  
Proper Shipping Name OCTANES

### IATA

UN-No UN1262  
Hazard Class 3  
Packing Group II  
Proper Shipping Name OCTANES

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------|-----------|------|-----|-------|------|------|-------|------|----------|
| Octane    | 203-892-1 | X    | X   | X     | X    | X    | X     | X    | KE-26612 |



# SAFETY DATA SHEET

n-Octane

Revision Date 31-Mar-2025

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

Health, Safety and Environmental Department

**Revision Date**

31-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**